Student Attendance Monitoring System

2. Problem Statement

2.1 In traditional systems, attendance is managed manually on paper which is time consuming and prone to errors.
2.2 It becomes difficult to maintain records for longer duration and generate accurate reports.
2.3 To overcome this, SAMS provides a digital solution to add students, mark attendance, save records and generate reports efficiently.
2.4 This ensures accuracy, reliability and reduces manual effort for teachers.

3. SDLC Model

3.1 Chosen model: Waterfall model

3.2 Justification

The requirements for the Student Attendance Monitoring System are clearly defined and unlikely to change frequently (e.g., mark attendance, add students, generate reports, etc.). Therefore, the Waterfall model is appropriate due to its sequential and structured approach, where each phase (Requirement analysis → Design → Implementation → Testing → Deployment) is completed before the next phase begins. This makes the project easier to manage, maintain, and ensures reliable functionality without much iterative development.

4. Requirements

4.1 Functional requirements

- Ability to add and delete students.
- Option to mark students as present or absent.
- Saving attendance session-wise with date, time and subject.
- View past records by filtering date, subject, and time.
- Generate monthly attendance reports with percentages.

4.2 Non-functional requirements

- Simple and user friendly interface.
- Reliable and accurate storage of records in csv files.
- Fast response time even with large number of students.
- Platform independent system working on python supported os.

- Error handling for invalid inputs, duplicate ids and missing data.

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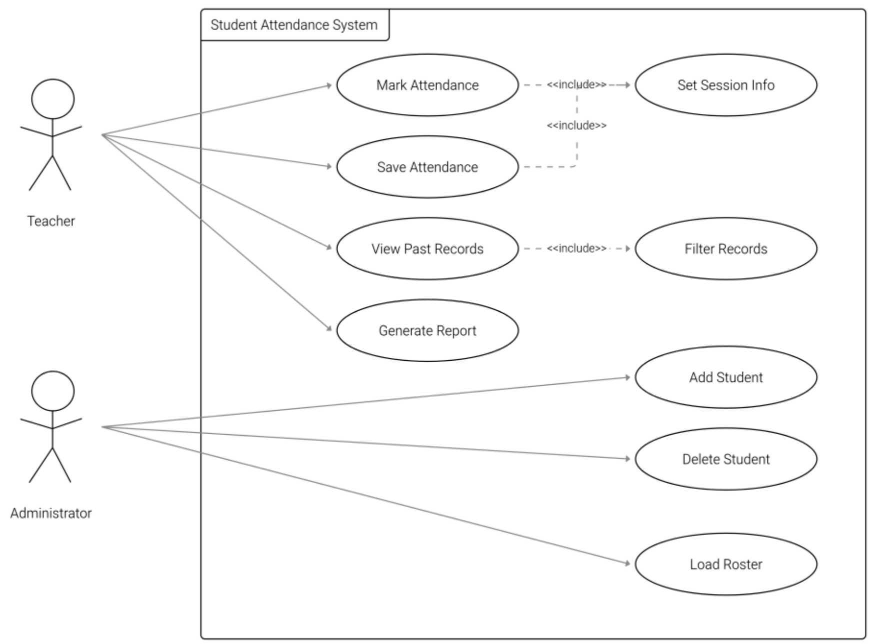
5. UML Diagrams

5.1 Use case diagram: The use case diagram illustrates the system's functional requirements, showing how the Teacher and Administrator interact with use cases like marking attendance, saving records, generating reports, and managing students.

Use Case Diagram Explanation

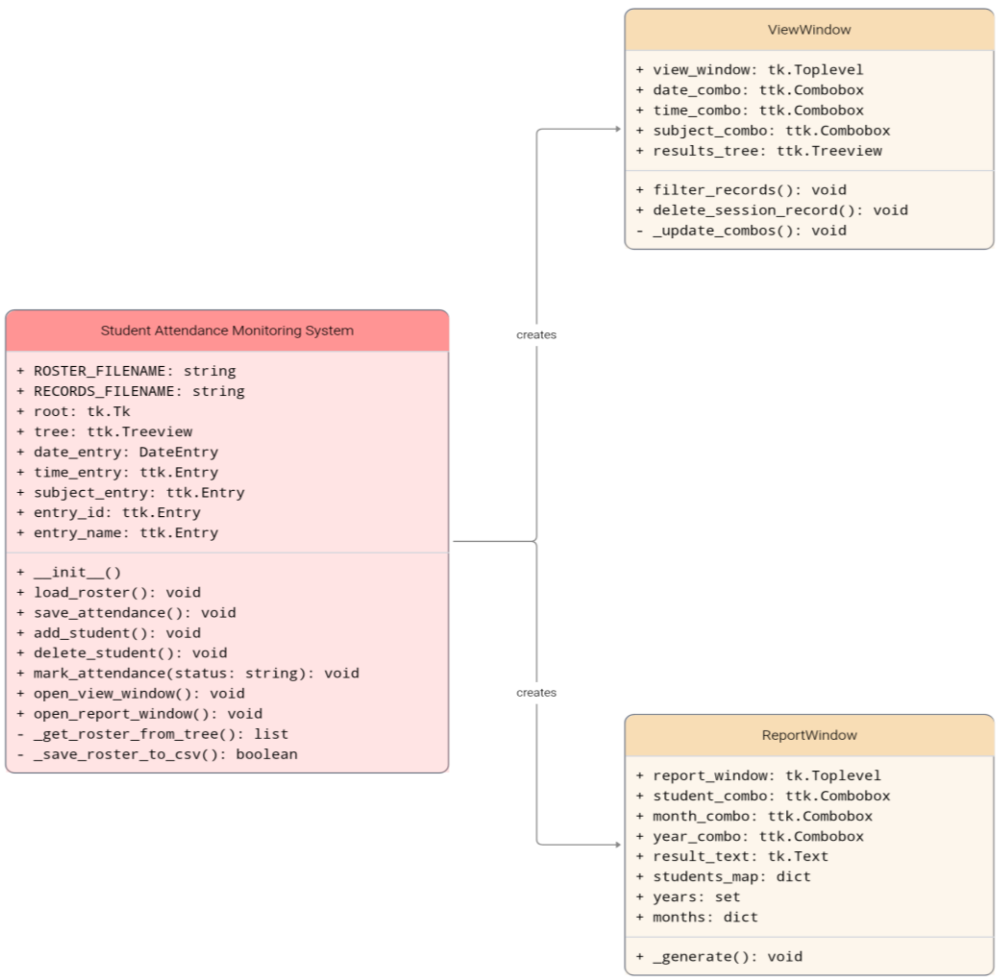
The use case diagram represents the functional requirements of the Student Attendance Monitoring System. The Teacher interacts with the system to mark, save, and view attendance, as well as generate reports. The Administrator manages the student roster by adding, deleting, and loading student details. Additional use cases such as setting session information and filtering records are included to support the main functionalities. This diagram clearly defines the roles and responsibilities of each actor in the system.



5.2 Class diagram: The class diagram shows the structural design of the Student Attendance Monitoring System, highlighting its main classes (Student Attendance Monitoring System, ViewWindow, ReportWindow), their attributes, and methods.

Class Diagram Explanation

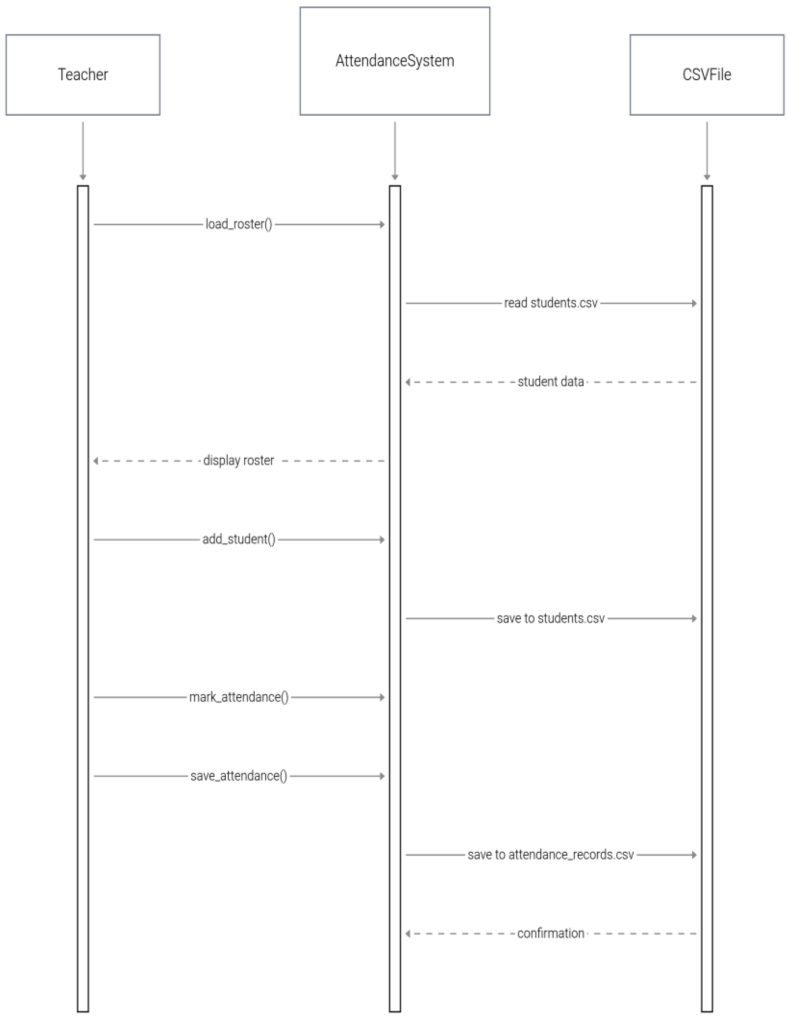
The class diagram depicts the object-oriented design of the system. The AttendanceSystem class manages core operations such as maintaining the roster, marking attendance, and saving records. It is associated with the ViewWindow and ReportWindow classes, which handle viewing past records and generating reports, respectively. The one-to-many relationships indicate that a single AttendanceSystem instance can create multiple ViewWindow and ReportWindow objects. This diagram provides a structured view of the system's components and their interactions.



5.3 Sequence Diagram: The sequence diagram shows the step-by-step interaction between the teacher, the system interface, and the database (CSV files) as students are added, attendance is marked, records are saved, and reports are generated.

Sequence Diagram Explanation

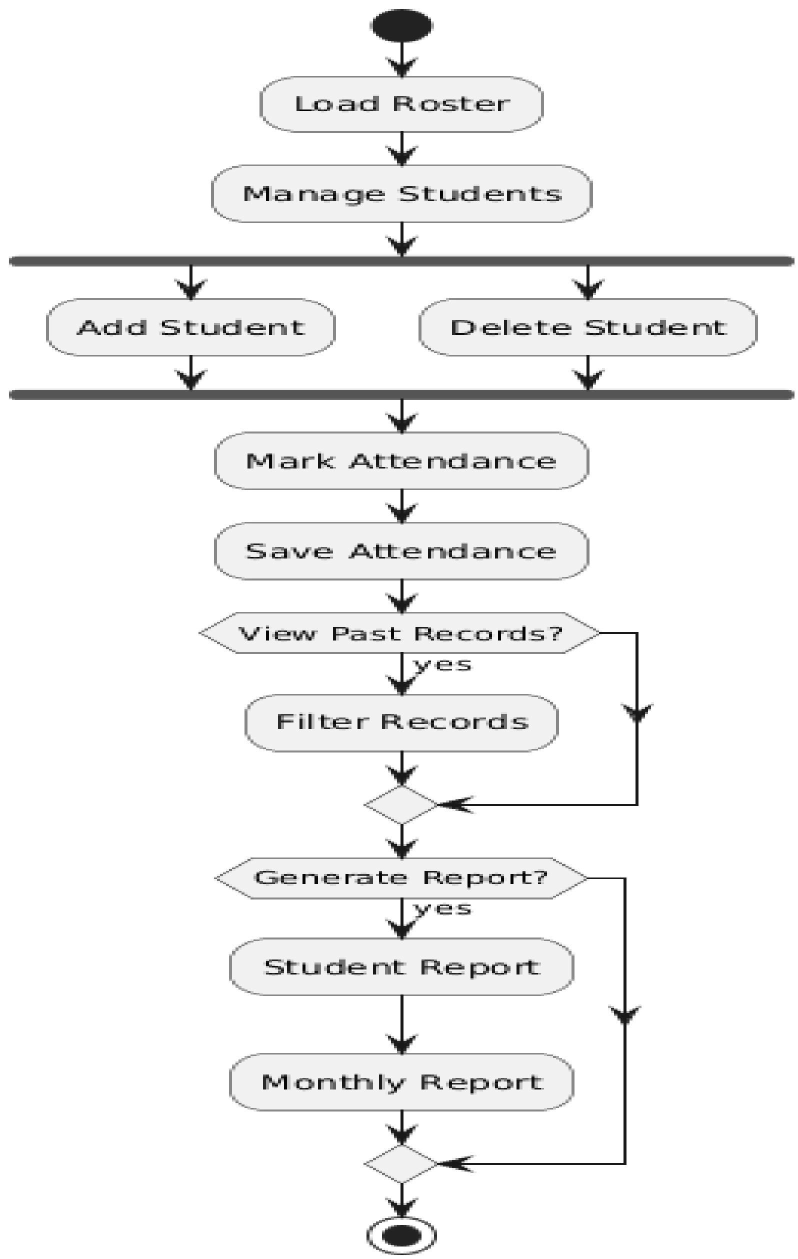
The sequence diagram illustrates the interaction between the Teacher, the Attendance System, and the CSV file storage. The Teacher initiates actions such as loading the roster, adding students, marking attendance, and saving attendance. The Attendance System processes these requests by reading from or writing to CSV files and provides confirmation or displays updated data back to the Teacher. This ensures a clear flow of control and data within the system.



5.4 Activity diagram: explains the workflow of adding student, marking attendance, saving and generating report.

Activity Diagram Explanation

The activity diagram illustrates the workflow of the Student Attendance Monitoring System. The process begins with loading the student roster, followed by managing students through options to add or delete entries. The teacher then proceeds to mark and save attendance. Decision nodes are used to check if the teacher wishes to view past records or generate reports. If chosen, records can be filtered and reports such as student or monthly summaries are produced. The diagram clearly represents the sequential flow of operations, decision-making points, and the overall functional behavior of the system.



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6. System Design

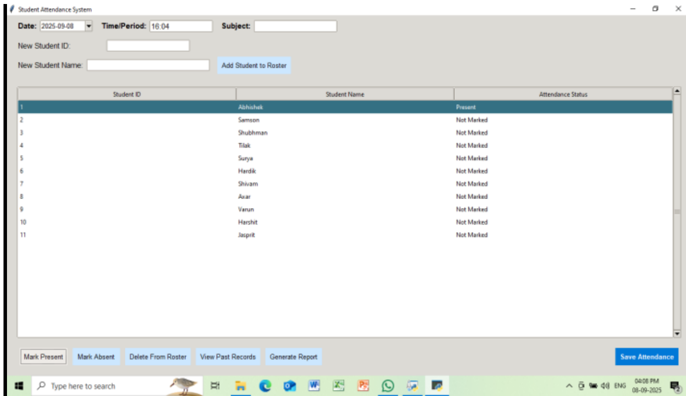
6.1 Gui layer: tkinter based interface for forms, buttons, and attendance tree view.
6.2 Data layer: csv files used for roster (students.csv) and attendance (attendance_records.csv).
6.3 Business logic layer: functions such as add_student(), delete_student(), mark_attendance(), save_attendance().
6.4 Flow of data: user input → gui interaction → data processing → csv storage → report generation.

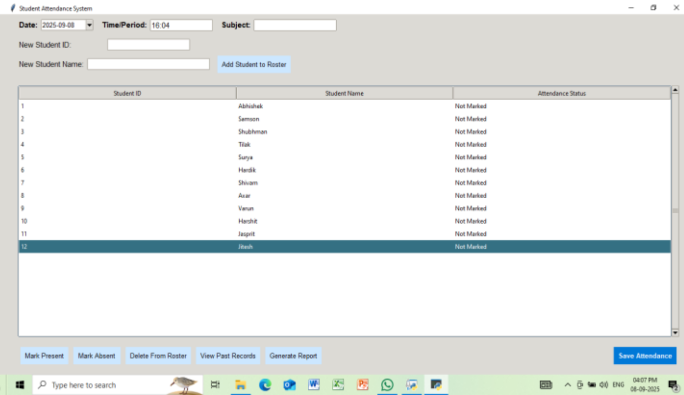
7. Testing Plan

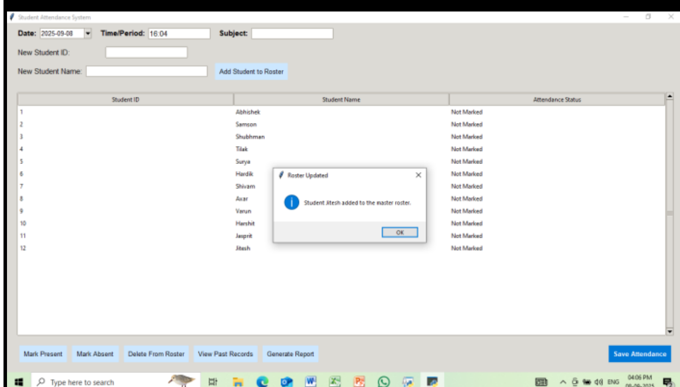
Test Case ID	Test Scenario	Input Data	Expected Output	Actual Output	Status
TC01	Add New Student	ID = "101", Name = "John Doe"	Student appears in the Treeview and is saved in `students.csv`	As expected	Pass
TC02	Mark Attendance as Present	Select student, click "Mark Present"	Status in the Treeview changes to "Present"	As expected	Pass
TC03	Save Attendance Session	Enter Date, Time, Subject → Click "Save"	Attendance data saved in `attendance_records.csv`	As expected	Pass
TC04	View Past Attendance Records	Select Date, Time, Subject → Click "View"	Display list of attendance records matching criteria	As expected	Pass
TC05	Delete Student from Roster	Select student → Click "Delete From Roster"	Student is removed from Treeview and `students.csv`	As expected	Pass
TC06	Generate Attendance Report	Select Student, Month, Year → Click "Generate Report"	Displays attendance percentage report in report window	As expected	Pass

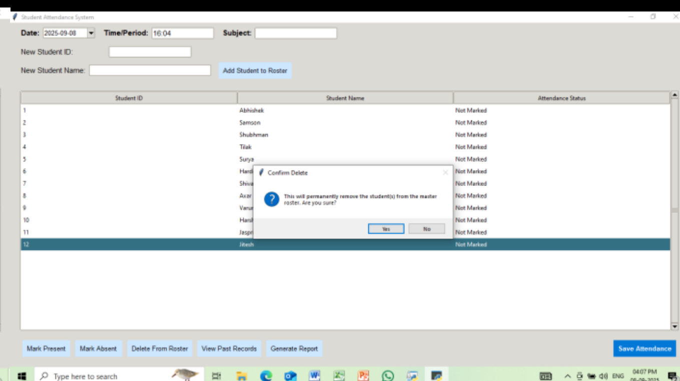
8. Screenshots Of Python Output

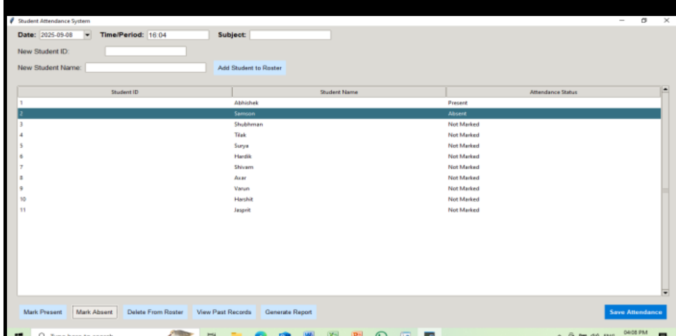
8.1 Main window showing session details and student input form.
8.2 Student roster list displaying ids, names, and attendance status.
8.3 Marked attendance view after updating status.
8.4 Past records window with filter options.
8.5 Report generation window with percentage calculations.

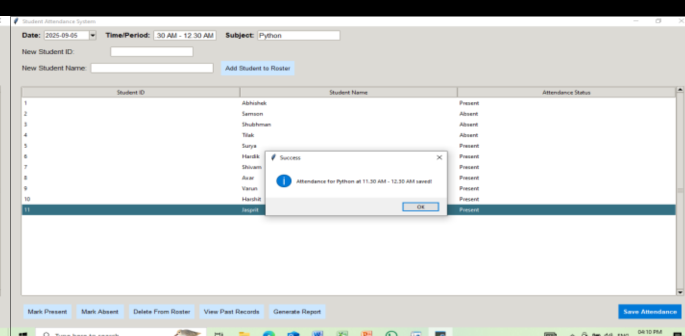

Student Attendance System
Date:
Time/Period:
Subject:
New Student ID:
New Student Name:
Add Student to Roster
Student ID
Student Name
Attendance Status
Mark Present
Mark Absent
Delete From Roster
View Past Records
Generate Report
Save Attendance

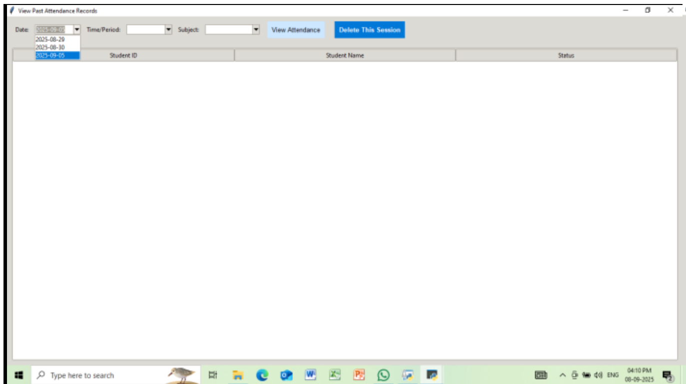

Student Attendance System
Add Student to Roster
Mark Present
Mark Absent
Delete From Roster
View Past Records
Generate Report
Save Attendance

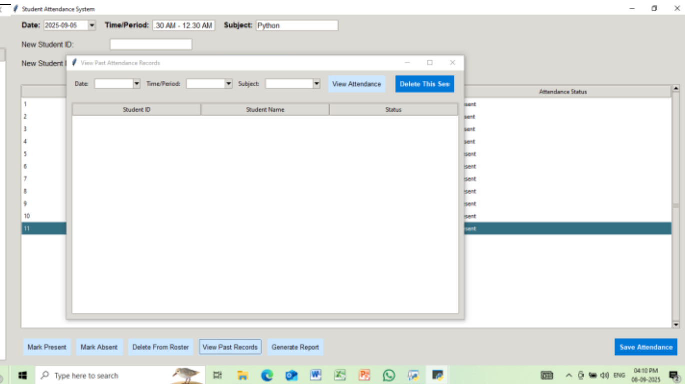

Roster Updated
Student Jitesh added to the master roster.
OK

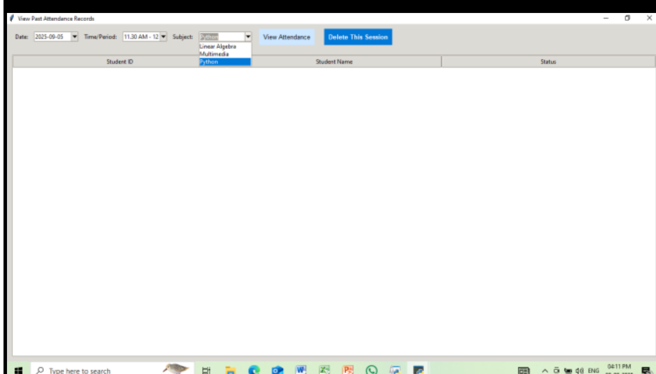

Confirm Delete
This will permanently remove the student(s) from the master roster. Are you sure?
Yes
No

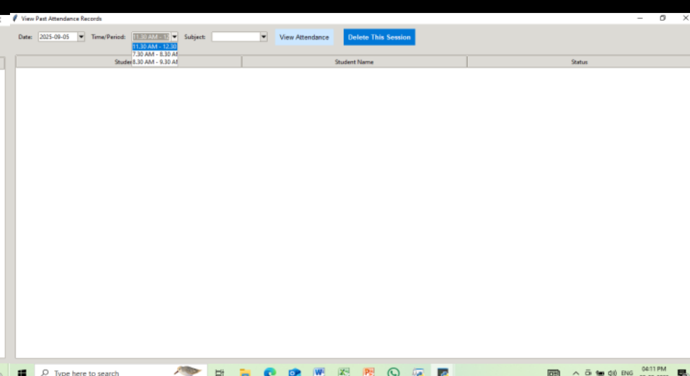

Student Attendance System
Mark Present
Mark Absent
Delete From Roster
View Past Records
Generate Report
Save Attendance

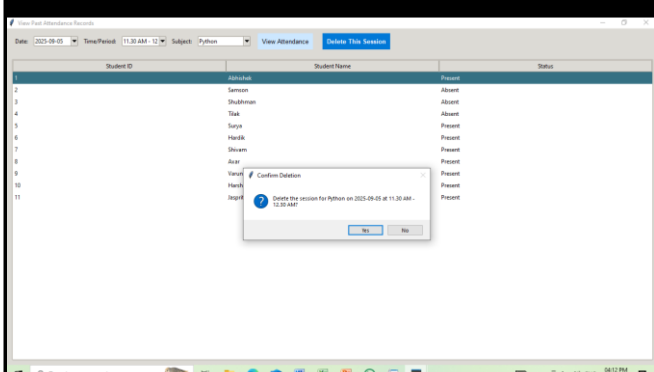

Success
Attendance for Python at 11:30 AM - 12:30 AM saved!
OK

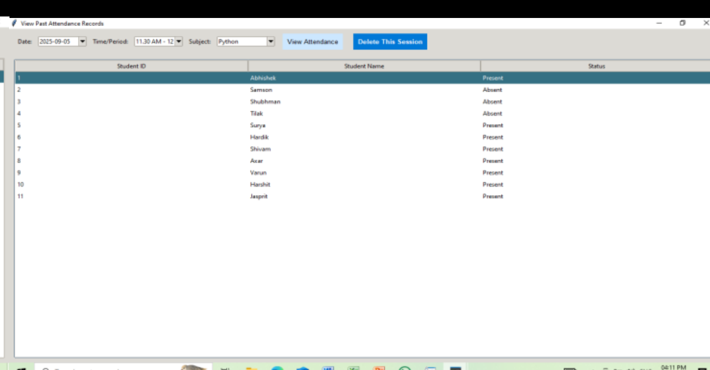

View Past Attendance Records
Date:
Time/Period:
Subject:
View Attendance
Delete This Session
Student ID
Student Name
Status

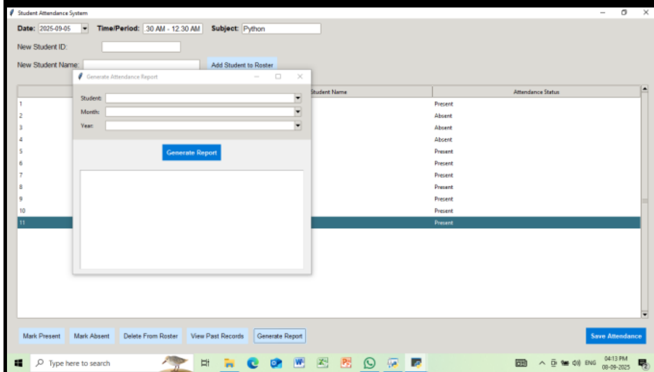

Student Attendance System
Date:
Time/Period:
Subject:
New Student ID:
Mark Present
Mark Absent
Delete From Roster
View Past Records
Generate Report
Save Attendance

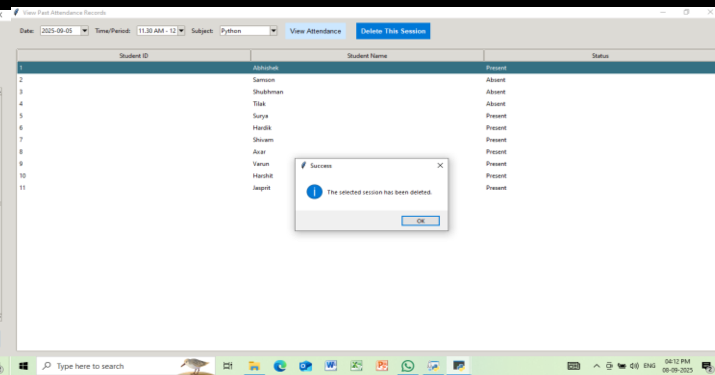

View Past Attendance Records
View Attendance
Delete This Session

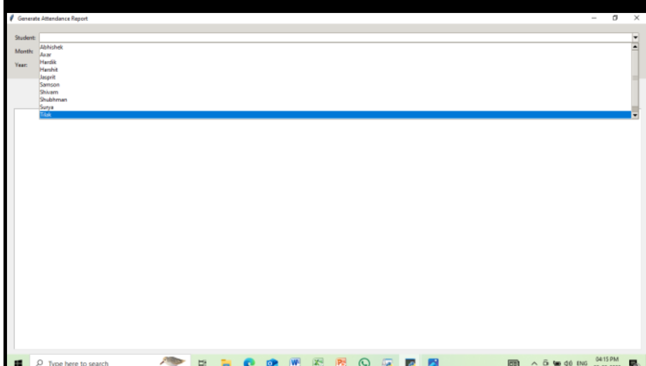

View Past Attendance Records
View Attendance
Delete This Session

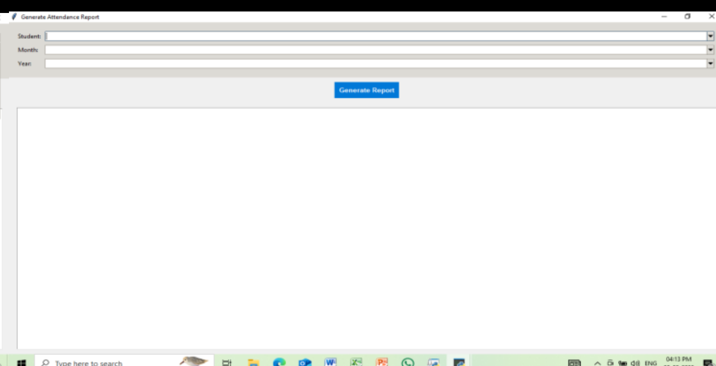

View Past Attendance Records
View Attendance
Delete This Session
Confirm Deletion

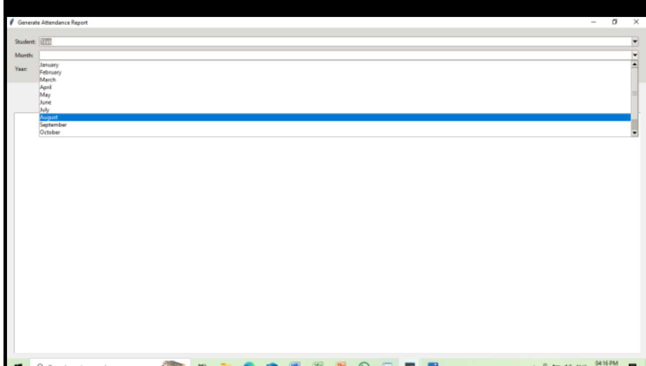

View Past Attendance Records
View Attendance
Delete This Session

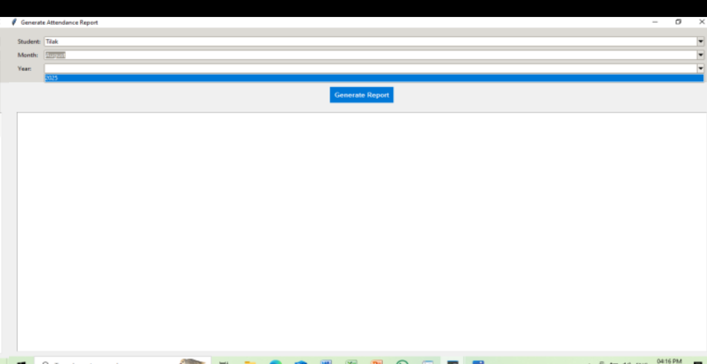

Student Attendance System
Generate Attendance Report
Generate Report
Mark Present
Mark Absent
Delete From Roster
View Past Records
Save Attendance

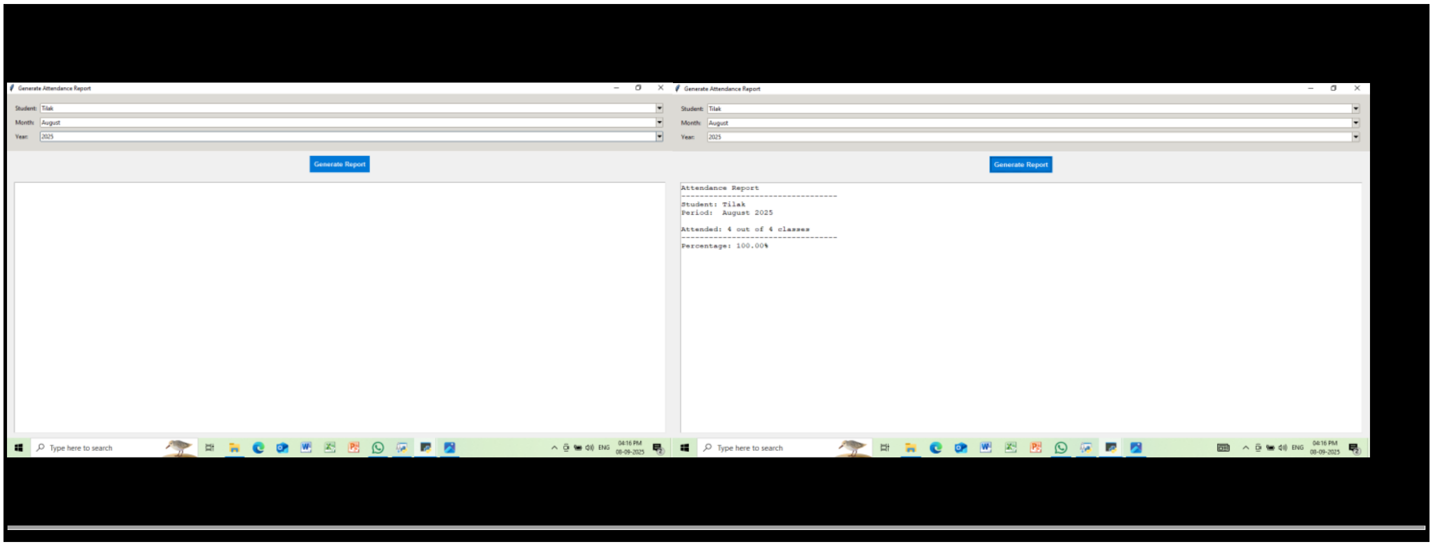

View Attendance
Delete This Session
Success
The selected session has been deleted.
OK


Generate Attendance Report


Generate Attendance Report
Generate Report


Generate Attendance Report


Generate Attendance Report
Generate Report



9. Conclusion And Future Scope

9.1 Conclusion: The SAMS automates attendance management, reduces errors, stores records securely, and allows easy report generation. It is lightweight, user friendly, and suitable for academic use.

9.2 Future scope:

- Integration with databases like mysql or sqlite.
- Cloud based access for multi-device usage.
- Biometric or rfid integration for automatic attendance.
- Mobile application support for android and ios.
- Exporting reports in excel or pdf format.
- Role based access for admin, faculty, and students.